Airbnb NYC 2019 Data Analysis

Mehak Badani

BSAN 360 – 002. FINAL PORJECT

Dataset Description

The dataset contains Airbnb listing data for New York City in 2019. It includes information about:

- Listing price
- Neighborhood and neighborhood group
- Room type (entire home, private room, shared room)
- Number of reviews and availability
- The dataset helps analyze pricing trends and listing distribution across NYC.

Data Preparation

- Cleaned missing or inconsistent values
- Removed irrelevant or duplicate records
- Converted variables into appropriate formats
- Prepared data for analysis and visualization

Research Question

Main Question:

How do location and property type affect Airbnb listing prices in NYC?

Supporting Questions:

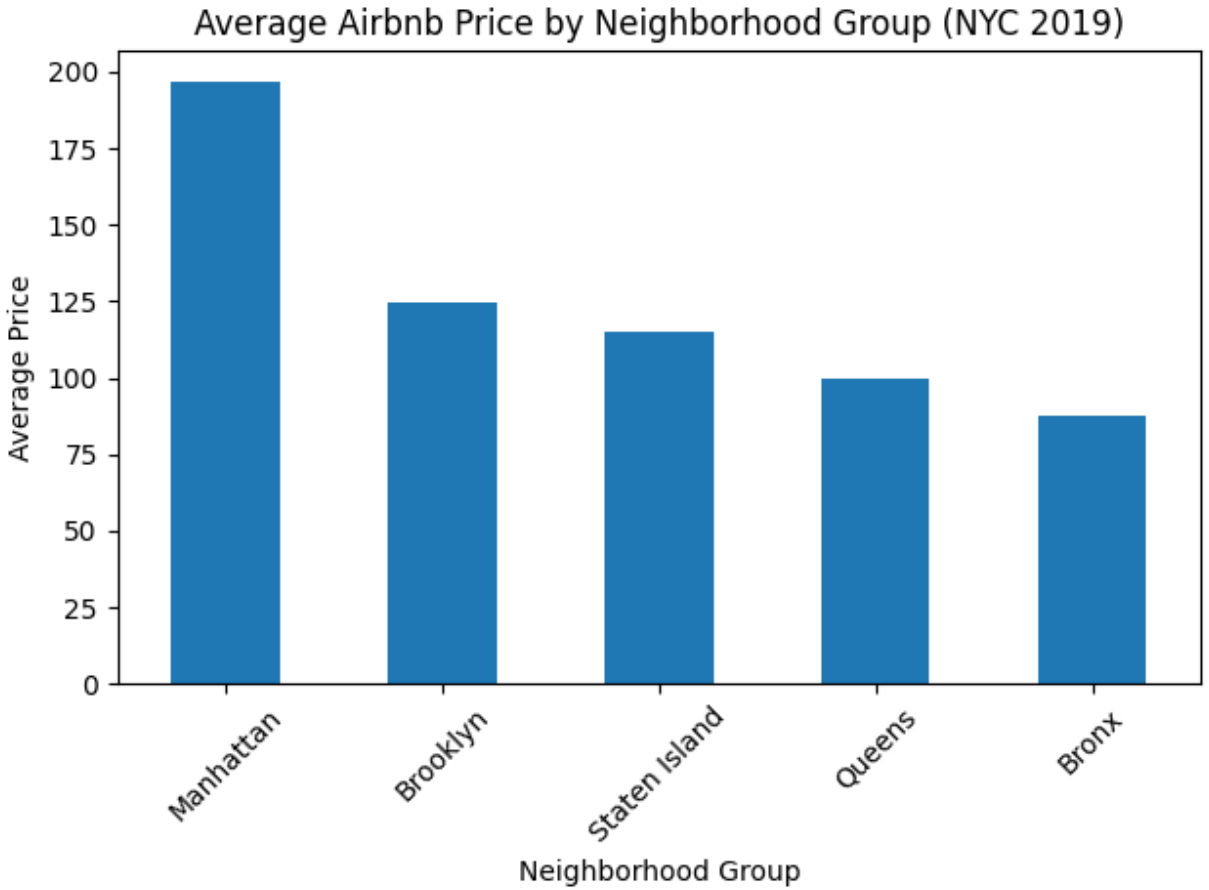
Which neighborhoods are most expensive?

Which room types generate higher prices?

Methodology

- Conducted exploratory data analysis (EDA)
- Grouped data by neighborhood and room type
- Used visualizations such as bar charts
- Compared price trends across different variables

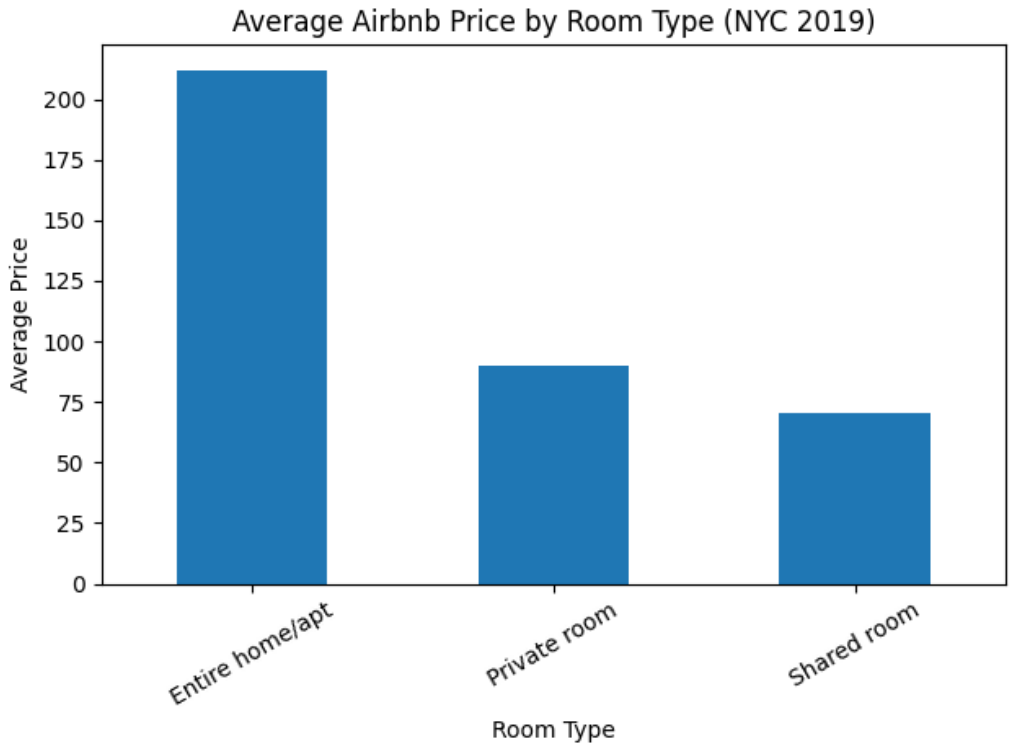
Price Distribution by Neighborhood Group



Manhattan has the highest average Airbnb prices
Brooklyn follows with moderately high prices
Queens and Staten Island have lower average prices
The **Bronx** has the most affordable listings.

This chart shows the average Airbnb prices across different neighborhood groups in New York City.
Manhattan clearly has the highest prices, followed by Brooklyn.
On the other hand, areas like Queens, Staten Island, and the Bronx are more affordable.
This shows that location plays a major role in determining Airbnb pricing

Price Comparison by Room Type



Entire home/apartment listings have the highest average price
Private rooms are significantly more affordable
Shared rooms have the lowest prices
This shows that room type strongly influences Airbnb pricing

This chart shows how Airbnb prices vary by room type.
Entire homes or apartments are the most expensive option, while private and shared rooms are much more affordable.
This indicates that the type of accommodation plays a major role in pricing differences

Key Insights

- Airbnb prices vary significantly by location, with Manhattan being the most expensive
- Room type strongly impacts pricing, with entire homes costing much more than shared spaces
- More affordable options are concentrated in areas like Queens and the Bronx
- Pricing patterns show that both location and accommodation type are key drivers of revenue

Managerial Implications

- Hosts in high-demand areas like Manhattan can maximize revenue through premium pricing
- Hosts in lower-priced areas can attract customers by offering competitive and affordable pricing
- Property owners should choose entire home listings to increase potential earnings
- Pricing strategies should consider both location and room type to optimize revenue

Summary

- The analysis explored Airbnb pricing patterns in New York City
- Location and room type were identified as key factors influencing price
- Manhattan and entire home listings showed the highest prices
- Data-driven insights can help hosts make better pricing decisions

In conclusion:

This analysis showed that both location and room type significantly influence Airbnb prices.

Understanding these factors can help hosts and businesses make more informed decisions

Thank You!